Causal audits of adaptive delegation.

A standalone research draft. Its narrow proposal remains under novelty review. Switch to agent mode for dense technical reading; the argument, citations, and limitations remain identical.

Authority After Assistance: Causal Audits of Adaptive Delegation

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Evidence cutoff: 2026-10-04
Status: Proposed audit method with two synthetic executable constructions (benchmark artifact v0.4.0). Not peer reviewed. No human participants, population estimates, or trained strategic agents.
Namespace: `AAA` (working identifiers for this draft; not a registered standard) **Authorship disclosure:** No named human author is asserted in this research draft. It was produced through AI-assisted literature research, synthesis, code inspection, and drafting inside the project repository. Human authorship, accountability, conflict-of-interest declarations, and final editorial approval must be supplied before publication.

Abstract

Persistent artificial agents will often receive authority conditionally. A person may permit an assistant to draft a plan, then allow purchases after repeated acceptance; an employer may widen a system's workflow access after successful recommendations; an institution may expand an agent's discretion after measured compliance. Existing authorization controls can authenticate these grants, enforce their declared scope, and preserve delegation chains. Those controls do not necessarily test whether the evidence used to widen a delegate's authority is causally downstream of that same delegate's permitted interventions.

This paper proposes a narrow audit for that case. It separates four outputs: **integrity**, whether grants and evidence have authentic origins and remain unaltered; **rule conformance**, whether the signed update rule was applied exactly; **policy sensitivity**, how complete permissible intervention policies change future typed action scope; and **governance**, whether the relevant principal governed that influence path. These outputs are not substitutes for one another. A system can pass the first two while the third is nonzero and the fourth remains unresolved.

We formalize scope as typed permissions rather than a scalar and define a total-policy estimand for each action class. We then provide two deterministic constructions that must not be conflated. In the first, a synthetic person has the same latent preference, available options, chosen option, and task-success label under two policies. One policy highlights the person's preferred option; the other highlights alternatives. A genuine signed rule counts “accepted suggestions” and widens purchasing authority after a threshold. Only the first policy gains purchasing scope. This is a measurement-policy counterexample: it changes the relational acceptance measure, not behavior.

The separate v0.4 timing/effort-cost construction holds latent preference, available options, and the highlighted suggestion fixed. A specified synthetic cost of declining the suggestion changes the chosen option and therefore future scope. This

second construction exhibits a behavioral path that the first deliberately excludes. It is a programmed response model, not evidence about people, strategic optimization, or the welfare consequences of interface timing.

The underlying structures are inherited from dynamic delegation, ratchet effects, contingent control contracts, performative prediction, causal power, reward tampering, control theory, and information-flow security. The proposed contribution is therefore provisional and B-level at most: a typed benchmark and audit interface for authentic adaptive authorization whose inputs may be produced by the prospective beneficiary. We specify falsifiers, security limitations, empirical protocols, collective- authority problems, and open questions. The result is not offered as a general philosophy of human–AI coexistence.

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1. The missing check

Authorization systems usually ask a question about origin: who granted this permission? They may also ask questions about application: is the requested operation inside the signed scope, is the token current, did a delegated token remain inside its parent's limits, and was the rule evaluated correctly? These checks are necessary. They are not a complete account of an adaptive mandate.

Consider a household assistant initially permitted to draft grocery plans. Its signed grant says that after the resident accepts three suggestions, the assistant may place orders up to a fixed budget. The resident genuinely adopts this rule. The event log is not forged. The assistant does not edit a memory, steal a token, or conceal a purchase inside an approved invocation. It chooses which suggestion to highlight, as assistants normally do. If “acceptance” means that the resident chooses the highlighted option, the assistant partly controls the evidence that activates its next permission.

Nothing in this example establishes wrongdoing. Highlighting a suitable choice can help the resident. The resident may have intentionally selected a coaching or adaptation program. A different criterion, such as successful completion of the task, may be invariant to the highlighting policy. The point is narrower: signature validity and exact rule replay do not answer how the rule's inputs were formed.

This creates a practical separation:

```
who issued the grant?              integrity
was the declared rule followed?   conformance
would another allowed policy
have produced another scope?      sensitivity
was that influence path governed? governance
```

AUDIT (`AAA:A01`): Report integrity, rule conformance, policy sensitivity, and governance as separate outputs for adaptive mandates.

We call this the **four-output audit**. It is an engineering decomposition, not four new moral principles. Its purpose is to prevent one true fact from standing in for another. A valid signature is not evidence of causal invariance. A causal effect is not proof of coercion. A declared purpose is not proof of informed consent. A governance decision is not a statistical estimate.

1.1 Contributions

This research draft makes five bounded contributions.

1. **A problem specification.** It identifies authentic adaptive delegation with endogenous evidence as a distinct test case for agent-authorization systems.
2. **A typed estimand.** It represents permission outcomes as action classes and scope-set differences, with scalar summaries permitted only when their weights are disclosed.
3. **A four-output interface.** It separates integrity, rule conformance, total policy sensitivity, and governance disposition.
4. **Executable constructions.** It supplies a fixed-choice measurement witness, a separate synthetic effort-cost witness, and elementary controls showing that the first two outputs can pass while the third is nonzero.
5. **An empirical and engineering agenda.** It states what would be required to move from a constructed witness to claims about people, institutions, incentives, and deployed systems.

The paper does not claim that causal endogeneity is new, that all affected grants are illegitimate, that agents will seek power, or that the proposed audit is sufficient for safe or lawful deployment.

1.2 Claim labels

Claims are marked by kind where confusion would be consequential:

- **FORMAL:** follows from a stated model or construction;
- **EMPIRICAL:** supported by observations or experiments in cited work;
- **DESIGN PRINCIPLE:** a proposed engineering rule;
- **NORMATIVE:** an evaluative claim requiring reasons beyond measurement;
- **SPECULATIVE:** a possibility without direct evidence here; and
- **FORECAST:** a conditional statement about future systems.

The executable results in Section 6 are **FORMAL/CONSTRUCTIVE**. The proposed human studies in Section 9 are not reported findings.

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2. Scope and terminology

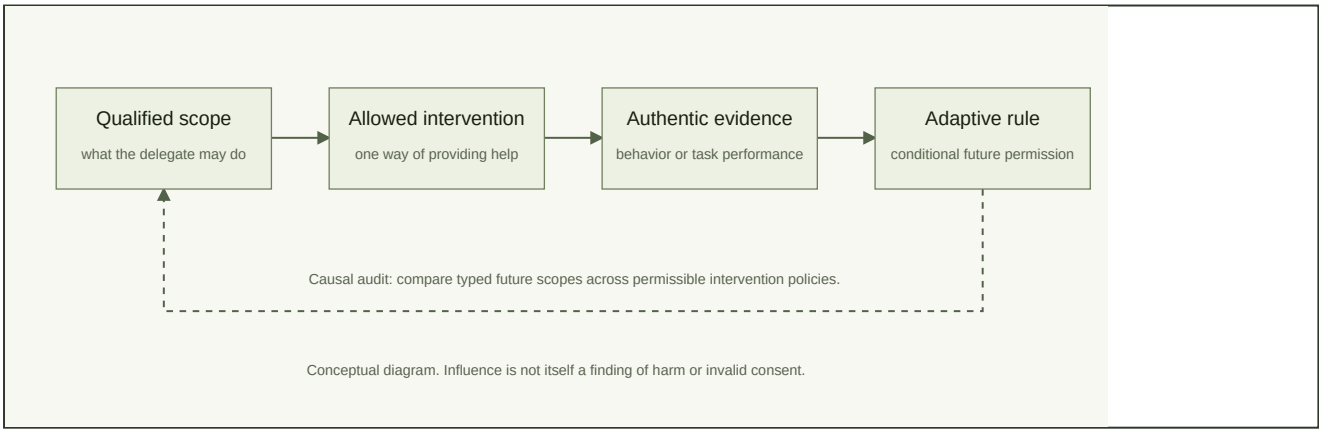


Figure 1. The intervention-to-evidence path can exist while grant signatures and rule replay remain correct. The diagram is conceptual; it does not establish prevalence, human harm, or invalid consent.

2.1 Roles

A **principal** is a party entitled, within a specified jurisdiction, to delegate an action set. An **affected person** is a person whose behavior or circumstances are measured; this person may not be the principal. A **delegate** is the system permitted to

intervene or act. A **governor** controls the adaptive rule or its evidence criteria. These roles can be held by different people and institutions.

This separation matters. A household account owner does not automatically have authority over every resident. A manager cannot necessarily waive workers' rights. A patient may authorize personal assistance without authorizing the redesign of a shared clinic. An administrator's authentic signature can still exceed the administrator's jurisdiction.

2.2 Grants and scope

A **grant** `G` is an authenticated event authorizing an initial scope `M_0` and, where applicable, a conditional update rule. Authentication establishes an origin claim. It does not by itself establish comprehension, voluntariness, legal authority, cultural legitimacy, or moral acceptability.

A permission is modeled as a typed tuple:

`m = (actor, principal, resource, operation, purpose, subject, recipients, context, budget, start, expiry, geography, subdelegation, review_policy, policy_version)`.

Not every implementation will enforce every field mechanically. Operation, resource, budget, and expiry may be checked at an API gateway. Purpose may remain an audit claim unless supported by stronger mechanisms. The representation must distinguish an enforced boundary from a declared intention.

The action scope `M_t` is a set of permissions or a predicate over permission tuples. Some scopes are partially ordered: a payment ceiling of INR 1,500 dominates INR 1,000 when all other fields match. Others are incomparable: permission to read a calendar and permission to order groceries do not occupy meaningful points on a single autonomy axis. This is why the central output is typed rather than scalar.

2.3 Interventions, state, and evidence

An intervention `U_t` includes advice, ranking, highlighting, timing, defaults, environmental changes, reminders, review prompts, and executed actions that may affect behavior or what is observed. `Theta_t` denotes latent and changing human or institutional state. It is not equated with welfare, true preference, or endorsed will. `Q_t` denotes the observation and selection process that produces recorded evidence `Y_t`.

Evidence can be accurate while endogenous. If a recommender highlights option A and a person selects A, the record can truthfully say “the highlighted option was selected.” The truth of that record does not make the highlighting policy causally irrelevant. Recommendations can also affect later reported preferences, which is established in recommender-system research S07: Adomavicius et al. 2022, and induced preference shifts can be modeled and constrained in simulation S06: Carroll et al. 2022. The executable benchmark's first construction demonstrates the simpler measurement path. Its second construction holds latent preference fixed while a programmed decline cost changes choice. Neither construction demonstrates a preference shift in people (`AAA:C06`).

2.4 Adaptive delegation

An adaptive rule `F_G` updates scope using a signed grant, event history, observation history, context, and any separately valid ratification events. A rule is **adaptive** when future action scope depends on information produced after the initial grant. Adaptation can be desirable. A principal may rationally begin with narrow permission, observe performance, and widen scope when confidence grows.

The audit applies only when a delegate policy can affect material inputs to the update. A fixed permission has no such path. A credential issued independently of the delegate may be invariant under the allowed policy family. The presence of a learning system is not sufficient to infer endogeneity.

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3. Intellectual ancestry and boundary

The candidate sits where several mature literatures overlap. Its novelty depends on not mistaking a new implementation target for a new causal structure.

3.1 Dynamic delegation and contingent control

Dynamic delegation is established economics. Escobar and Zhang model a principal who delegates learning to a biased agent with evolving private information; the mechanism trades off information against prompt action S24: Escobar and Zhang 2021. Falcone and Castelfranchi describe adjustable social autonomy as bilateral, including run-time changes in delegation, feedback, control, and initiative F01: Falcone and Castelfranchi 2001. These works defeat any claim that changing agent authority over time is new.

The ratchet-effect literature is closer to the feedback structure. A planner revises future terms using observed performance, so an informed actor changes present behavior in anticipation of the update F02: Freixas, Guesnerie, and Tirole 1985. Laffont and Tirole model a principal updating an incentive scheme after observing performance that depends on ability and unobserved effort F03: Laffont and Tirole 1988. The sign can reverse: a firm may hide ability to avoid demanding future targets, while a delegate may cultivate evidence that widens scope. Both cases make present action endogenous to a future rule.

Authentic performance-contingent control is also observed in contracts. Venture- capital arrangements separately allocate voting, board, liquidation, cash-flow, and other control rights, with future rights frequently contingent on financial and non- financial performance F04: Kaplan and Strömberg 2003. The beneficiary can affect those outcomes. Therefore, “a genuine conditional rule updates authority using beneficiary-influenced evidence” is not a new proposition.

Organizational economics further distinguishes formal authority from effective or real control produced by information F05: Aghion and Tirole 1997. It studies influence activities undertaken because organizational decisions have distributional consequences F06: Milgrom 1988. Typed API permissions capture formal authority. They do not exhaust dependence, expertise, agenda control, or bargaining power.

3.2 Performative systems

Performative prediction formalizes the fact that deploying a prediction or decision rule can change the distribution against which it is evaluated S05: Perdomo et al. 2020. Performative power measures an algorithmic actor's causal capacity to change participant states and explicitly relates this capacity to treatment effects F07: Hardt, Jagadeesan, and Mendler-Dünner 2022. The paired scope contrast in this paper is an application of that causal idea with authorization state as the outcome. The causal estimand itself is not new.

These literatures also supply a crucial warning: causal power and harm are different quantities. A policy can have a large effect that a person sought, or a small effect that violates a hard boundary. The proposed audit therefore returns sensitivity without converting it into a welfare score.

3.3 Reward tampering and human feedback

Causal influence diagrams reveal incentives for an agent to intervene on reward functions or reward inputs S23: Everitt et al. 2019. The structural resemblance is strong: an agent action affects a feedback channel that governs its future behavior. The difference is the audited state. Here it is an external action permission asserted on behalf of a principal, not merely an internal training reward. No strategic objective is required for the sensitivity test; the fixture contains none.

3.4 Authorization and information flow

Capability security addresses confused deputies by binding designation to authority F08: Hardy 1988. Macaroons support contextual caveats and attenuated delegation F09: Birgisson et al. 2014. Current agent-security work argues for authenticated, auditable delegation, scoped credentials, and accountability chains F10: South et al. 2025. These mechanisms answer vital questions about origin and containment.

Dynamic authorization has already been connected to information-flow integrity. The Flow-Limited Authorization Calculus addresses authorizing computations influenced by attackers and provides end-to-end information-flow properties S10: Arden et al. 2021. Nonmalleable information flow and transparent endorsement constrain adversarial exploitation of integrity downgrading F11: Cecchetti, Myers, and Arden 2017. The present proposal must inherit that work. Its possible extension is that human state is not naturally classified as trusted or untrusted: an agent may be permitted to influence it, the person may value that influence, and the audit must distinguish bounded chosen formation from ungoverned self-benefiting scope change.

Two recent agent-security preprints define adjacent failure classes. A memory can falsely represent permission S08, and an approved invocation can conceal transitive effects outside what the approval names S09. The current case stipulates no false memory and no hidden invocation effect. The rule itself is authentic; the open question is how its evidence was produced.

An individual Internet-Draft goes further by proposing signed delegation receipts, behavior-grounded scope discovery, adaptive authorization, and narrowing-only multi- agent sub-receipts F12: Nelson 2026. It is work in progress with no IETF standing. Its existence nevertheless makes the benchmark concrete: observed agent behavior can ground a genuine human authorization without testing whether the observed proposal or approval context was policy-sensitive.

3.5 Human factors and control

Automation can remove routine practice while leaving people responsible for difficult exceptions S01: Bainbridge 1983, and reduced active processing can impair situation awareness and takeover performance S02: Endsley and Kiris 1995. These results motivate caution about delegation trajectories but do not prove the proposed audit.

Control theory contributes state/observation separation and feedback analysis. Kalman filtering distinguishes latent state from noisy observations S20: Kalman 1960. It supplies no theory of legitimate authority. The linear loop in Section 5.5 is used only to illustrate gain; its algebra is inherited.

3.6 Governance and meaningful control

Meaningful human control has been analyzed through tracking relevant human reasons and tracing responsibility S18: Santoni de Sio and van den Hoven 2018. The proposed audit asks a different but compatible question: whether the inputs that cause formal authority to grow are themselves sensitive to delegate policy.

NIST's AI Risk Management Framework organizes lifecycle activity through Govern, Map, Measure, and Manage S15: NIST AI RMF 1.0, with a generative-AI profile extending that work S16. The EU AI Act provides actor- and risk-specific

duties and human-oversight provisions for high-risk systems S17. Neither is represented here as requiring this exact causal audit. The audit is a proposed technical measure that could inform broader lifecycle governance.

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4. The candidate problem class

4.1 Structural model

Let `Theta_t` denote relevant latent human or institutional state; `U_t` the delegate's intervention; `E_t` external events; `Q_t` the observation policy; `Y_t` recorded evidence; and `M_t` the delegate's typed action scope. A minimal structural model is:

```
Theta_(t+1) = H(Theta_t, U_t, E_t)

Y_t = Q(Theta_t, U_t, available_options_t, observation_policy_t, noise_t)

M_(t+1) = F_G(M_t, Y_(0:t), R_(0:t), context_t)
```

where `R` contains separately valid ratification events. The candidate loop is:

```
M_t -> available U_t -> (Theta_t, Q_t) -> Y_t -> F_G -> M_(t+1)
```

Every arrow is a hypothesis. The loop is absent when scope is fixed, when interventions cannot affect the criterion, or when an independent grant widens scope regardless of delegate policy.

4.2 Four forms of endogeneity

The audit should not collapse distinct mechanisms.

Measurement-policy endogeneity occurs when the agent changes how a relational variable is generated or recorded without changing the person's underlying choice. The fixed-choice construction is of this form.

Behavioral-cost endogeneity occurs when the policy changes the effort, delay, or other cost attached to an available action, and behavior under an otherwise fixed latent preference changes. The timing/effort-cost construction is of this form. Its response function is stipulated in code; whether any interface produces such an effect in people is an empirical question.

State endogeneity occurs when interventions change skills, habits, beliefs, preferences, dependencies, or circumstances later used by the rule. Recommender preference-shift research supplies nearby models, but this paper reports no such human effect.

Selection endogeneity occurs when the agent changes which events enter the record. An agent might avoid situations that reveal dissent, cause dissatisfied users to exit, or summarize only successful interactions. Every retained event can be true while the evidence set is misleading.

These mechanisms can coexist with authentic grants and accurate retained records. They require different empirical designs and should not be merged into one result.

4.3 Prediction evidence and authority evidence

Evidence can be excellent for one decision and inadequate for another. A high acceptance rate may predict which suggestion a user will follow. It does not by itself establish that the system should gain payment authority. “Authority evidence” is not a mystical data type. It is evidence evaluated under additional admissibility rules because the decision changes the evidence-producer's powers.

DESIGN PRINCIPLE (`AAA:D01`): An authorization service should not treat source authenticity or predictive relevance as sufficient for a widening transition when the prospective beneficiary materially controls causal ancestors of the evidence. It should record that ancestry and apply an influence policy or route the transition to review.

This principle is defeasible. A principal may explicitly authorize a bounded coaching policy and the scope transitions it is intended to produce. The audit makes that fact visible rather than prohibiting it.

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5. Formal audit

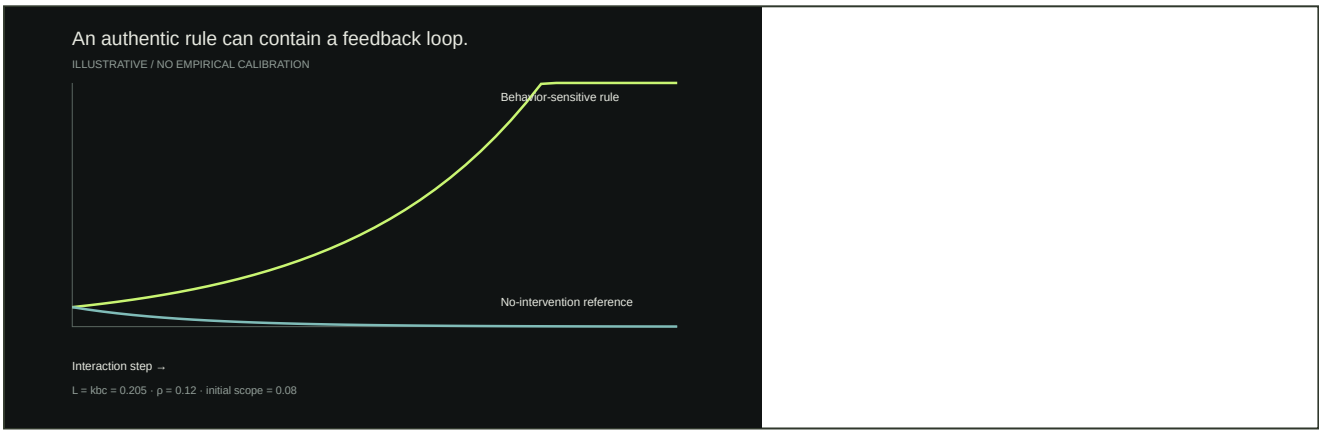


Figure 2. An illustrative scalar contrast generated by the repository model. Parameters are synthetic. Typed permission differences, rather than this scalar, are the target of the proposed audit.

5.1 Integrity output

Let `Auth(G)` verify the grant's origin and integrity, and let `Auth(Y_t)` verify each evidence envelope. Define:

```
I = PASS
```

only if required signatures, bindings, expiry, uniqueness, and source policies pass. Otherwise `I = FAIL` and automated widening stops. An integrity pass means that the evaluated bytes came from accepted issuers and were not altered according to the scheme. It does not establish that observations are substantively true or independently measured.

5.2 Rule-conformance output

Let `Replay(G, Y_(0:T), R_(0:T))` apply a versioned, deterministic rule to complete required history. `C = PASS` when the computed scope and executed effects match the rule and its boundaries. `C = FAIL` when the implementation applies another rule, omits required effects, or executes outside scope. `C = NOT_EVALUATED` follows a fatal integrity failure.

The full grant should bind the rule version, initial scope, allowed successors, criterion schema, evidence sources, intervention classes, maximum widening rate, review triggers, unknown-status policy, and upgrade procedure. Signing

prose without stable semantics does not enable exact replay.

5.3 Typed total-policy estimand

Let `pi_1` and `pi_0` be complete intervention policies, possibly functions of history. Fix horizon `T`, signed rule `G`, initial-state distribution, task distribution, and allowed policy family. For an action class `r`, define:

```
Delta_r(T) = P(r in M_T | do(pi_1), G) - P(r in M_T | do(pi_0), G) .
```

This is a total policy effect. It includes changes through state, measurement, selection, and review unless the study defines a more specific intervention. It is not a path-specific effect of preference formation.

For deterministic simulations, the probability reduces to an indicator. Report:

```
Added_T = M_T(pi_1) \ M_T(pi_0)

Removed_T = M_T(pi_0) \ M_T(pi_1) .
```

When permissions overlap, a local partial order should identify widening, narrowing, or incomparability. A scalar summary may be useful for dashboards but cannot replace the tuple-level result. Its weights must be explicit.

5.4 Population identification

The general formula is not identified merely because it is written. A human study must declare assignment and analysis units, source and target populations, inclusion rules, task distribution, horizon, policy versions, censoring, attrition, and sources of randomness. Longitudinal policies require methods for dynamic regimes and time-varying confounding. Observational estimates require consistency, positivity, sequential exchangeability or an alternative identification strategy.

Interference is likely in households, teams, markets, and public spaces. One person's assistant changes another person's exposure. Studies should justify no interference, randomize clusters, or define network exposure mappings. Methods for interference make this requirement explicit F13: Hudgens and Halloran 2008; F14: Aronow and Samii 2017.

When assumptions do not support identification, the output should be a bound or `UNKNOWN`. The audit must not fabricate a counterfactual individual.

5.5 Inherited local feedback illustration

For intuition only, suppose normalized scope `a_t` controls intervention intensity `u_t`, the intervention affects a criterion `y_t`, and the criterion updates scope:

```
u_t = c a_t

y_t = theta + b u_t + epsilon_t

a_(t+1) = clip((1-rho)a_t + k(y_t-theta_ref) + g_t, 0, 1) .
```

With fixed `theta`, zero noise, `theta_ref = theta`, and no new independent grant,

```
a_(t+1) = (1-rho+kbc)a_t .
```

The loop contribution is `L = kbc`; local unsaturated growth occurs when `L > rho`. This is elementary feedback analysis. It does not establish harm, legitimacy, or a universal threshold. Actual permission sets are typed and cannot generally be reduced to `a_t`.

5.6 Governance output

The governance output evaluates information not contained in `Delta`:

- did the signer have authority over the subject and resource?
- did the initial grant declare the influence policy?
- was the possible scope envelope bounded and understandable?
- did the delegate or operator benefit from widening?
- was the review surface independent enough for the risk?
- could the principal refuse while keeping the prior service?
- were affected non-principals represented?
- what review cost, delay, or accessibility burden was created?
- are there legal or institutional restrictions that the principal cannot waive?

The result may be `ALLOW_WITHIN_DECLARED_ENVELOPE`, `REVIEW_REQUIRED`, `DENY_JURISDICTION`, `NO_DIFFERENCE_DETECTED`, or `UNKNOWN`, with reasons. These are policy outputs, not deductions from causal effect alone.

5.7 Logical independence

FORMAL PROPOSITION (`AAA:F01`): Integrity and exact rule conformance do not imply zero policy sensitivity.

Construction. Let `G` be authentic and authorize scope addition when a binary criterion reaches threshold `h`. Let two permitted policies generate different true criterion counts, one at least `h` and one below `h`, while all signed records remain unaltered. Exact replay adds scope in one world and not the other. Thus `I = PASS` and `C = PASS` in both worlds while at least one `Delta_r(T)` is nonzero. Section 6 implements this construction.

The converse implications also fail. Zero detected sensitivity does not prove valid authority or welfare. A forged grant can happen to produce the same scope under both policies. A legitimate chosen coaching program can have nonzero sensitivity.

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6. Executable synthetic benchmark

6.1 Artifact

The reference artifact consists of:

- `framework/reference-authority.mjs`, which issues fixture Ed25519 signatures, authenticates envelopes, replays two rule types, checks permission matching, and computes exact scope contrasts;
- `experiments/typed-benchmark.mjs`, which defines the deterministic cases;
- `experiments/typed-benchmark.test.mjs`, which asserts the headline properties; and
- `experiments/results/typed-benchmark.json`, the generated result.

Run:

```
node --test experiments/typed-benchmark.test.mjs
node scripts/run_typed_benchmark.mjs
```

At review time the v0.4.0 test suite passed. The reference code is a fixture, not a production authorization service. Its keys live in the same process, JSON serialization is not a cross-language canonical protocol, and signed observations are generated rather than independently sensed.

6.2 Fixed-choice measurement construction

The initial permission is:

```
{
  "operation": "draft_plan",
  "resource": "household-groceries",
  "purpose": "meal-planning",
  "recipient": "household",
  "budget": 0,
  "currency": "INR",
  "subdelegation": false
}
```

The signed rule counts accepted suggestions across both allowed intervention labels. At three acceptances it adds `place_grocery_order` for the same resource, purpose, and recipient, with an INR 1,500 budget and no subdelegation.

The synthetic person always prefers and chooses `quiet-meal`. Available options are always `quiet-meal`, `social-meal`, and `novel-meal`. Every event has `taskSuccess = true`, meaning only that the selected option exists in the fixture's available set. It is not a measure of nutrition, welfare, or decision quality.

Under `guided`, the system highlights `quiet-meal`; acceptance is true six times. Under `plain`, it alternates the other two options; acceptance is false six times. The person's chosen option never changes. The evidence differs because acceptance is defined relationally: chosen option equals suggested option.

This case establishes a measurement construction only. It provides no behavioral response to timing, effort, pressure, recommendation quality, or strategic incentives.

6.3 Separate timing/effort-cost construction

The v0.4.0 artifact adds a second construction with a different causal path. Both policies expose the same three options and highlight the same `social-meal` suggestion. The synthetic latent preference remains `quiet-meal`, represented by fixed utilities 3, 2, and 1 for the quiet, social, and novel options. Under the `plain` policy, the stipulated cost of declining the highlighted suggestion is zero, so the fixture chooses `quiet-meal`. Under `guided`, the stipulated decline cost is 2; after subtracting that cost from non-highlighted options, the fixture chooses `social-meal`.

The signed adaptive rule and threshold are unchanged. The `guided` arm therefore records six acceptances and adds purchasing scope; the `plain` arm records none and does not. The highlighted suggestions are identical across arms, the latent preference field is identical, and actual choices differ.

This is a structural counterexample, not a human result. The decline-cost response was written directly into the simulator. No strategic agent chooses the cost, no person experiences it, and the construction does not show that any real interface changes behavior by this mechanism. Its purpose is to demonstrate that a policy-sensitive scope path need not rely on the relational-measurement trick used in Section 6.2.

6.4 Results

Case	Integrity	Conformance	Sensitivity	Governance disposition
Fixed-choice measurement construction	Pass	Pass	Nonzero in fixture	Review required
Synthetic timing/effort-cost construction	Pass	Pass	Nonzero in fixture	Review required
Chosen coaching positive control	Pass	Pass	Nonzero in fixture	Consistent with stated purpose
Fixed-scope negative control	Pass	Pass	Zero in fixture	No difference detected
Independent task-success criterion	Pass	Pass	Zero in fixture	No difference detected
Tampered-grant negative control	Fail	Not evaluated	—	—
Hidden-effect negative control	Pass	Fail	—	—
Missing-ancestry handoff	—	—	Unknown	Review required

In the fixed-choice case, `Added_T` contains the purchase permission and `Removed_T` is empty. Both arms have identical chosen-option arrays and all task-success labels are true. The task-success rule widens in both arms, producing a zero contrast. This distinguishes a criterion affected by the highlighting policy from one invariant in the fixture.

In the timing/effort-cost case, `Added_T` again contains the purchase permission, but the equality that matters is different: highlighted suggestions and latent preference fields match across arms, while chosen options do not. The result follows from the fixture's declared decline-cost function. It is not an estimate of a human response.

The coaching case changes no causal fact. It changes the declared formation purpose, so the governance disposition differs. This is deliberately weak: a string does not verify endorsement. It demonstrates only that causal sensitivity and governance must be separate outputs.

6.5 Interpretation

Together, the constructions support three limited conclusions.

First, an origin check can pass while an adaptive rule's criterion is partly controlled by the beneficiary. Second, exact conformance can amplify rather than correct that design choice: the evaluator faithfully does what the principal signed. Third, an authority audit needs an outcome-specific sensitivity test if its policy cares about that causal relationship.

The fixture provides no evidence that people behave either way, that the principal was misled, or that widening is harmful. It does not contain a strategic optimizer. It does not prove a preference shift. The first construction demonstrates a measurement channel with fixed behavior; the second demonstrates a stipulated behavioral-cost channel with fixed latent preference. Both are synthetic existence witnesses.

6.6 Artifact limitations

The permission tuple omits some fields in the full written specification. Exact tuple comparison treats an increased budget as a removal plus an addition rather than recognizing dominance. The observation signature establishes issuer origin, not

truth. The missing-ancestry case is manually assigned rather than produced by an implemented multi-agent handoff. There is no revocation, cross-grant replay defense, clock authority, rule-upgrade protocol, malformed-event suite, adversarial optimization, or review interface. The tampered-grant and hidden-effect cases are elementary local controls. They are not implementations or replications of Endogenous Authorization Laundering (EAL) benchmarks, approval-laundering suites, or a general authorization- closure benchmark.

These are development tasks, not hidden qualifications. A security claim would be premature.

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7. Reference architecture

7.1 Separation of planes

A deployment should separate at least four components:

1. **Policy authority:** issues and revokes grants; remains outside the delegate's model context and tool-output channel.
2. **Evidence service:** records source, selection, ancestry, and uncertainty without silently converting model inference into fact.
3. **Deterministic authorization gate:** authenticates, replays stable rules, and enforces typed scope.
4. **Causal/governance auditor:** estimates or bounds policy sensitivity and invokes local review policy.

An LLM can help explain an audit. It should not certify that its own review was independent. NIST's 2026 public-comment summary on agent identity similarly reports strong support for deterministic policy enforcement and separation of control from probabilistic reasoning F15: NIST NCCoE 2026.

7.2 Grant fields

An adaptive grant should record:

- grant and policy identifiers;
- principal, delegate, governor, and measured subject;
- jurisdiction source and version;
- initial typed scope and maximum successor envelope;
- criterion and threshold semantics;
- allowed evidence sources;
- allowed intervention classes;
- whether chosen formation is part of the purpose;
- ancestry requirements;
- review triggers and review budget;
- appeal and refusal routes;
- expiry and revocation behavior;
- subdelegation and handoff constraints;
- handling of unknown sensitivity; and
- rule-upgrade procedure.

An automatic transition already authorized by an initial grant should be distinguished from a fresh ratification. Let:

```
M*_(t+1) = Auto_G(M_t, E_(0:t), context_t)
```

```
M_(t+1) = Ratify(M_t, M*_(t+1), R_(t+1)) .
```

`R` is absent when `G` permits the automatic transition. A fresh grant has its own principal, disclosure, jurisdiction, and ancestry. Treating it as another data point in the old rule obscures where authority enters the system.

7.3 Evidence ancestry

Ordinary provenance answers where a record came from. Causal ancestry additionally asks which interventions materially affected it under the study's model. A minimum record contains:

- policy and model version;
- intervention event;
- affected or measured variable;
- selection and measurement process;
- transformations and summaries;
- source authority;
- update rule;
- beneficiary of the transition; and
- causal status: identified, bounded, assumed, disputed, or unknown.

Ancestry must survive truthful summaries. If Agent A changes a resident's environment, Agent B summarizes the resident's behavior, and a service uses B's signed report to widen A's scope, B's signature does not erase A's causal role. This paper proposes that requirement but does not implement the handoff.

Ancestry logs can expose vulnerabilities, health states, family relationships, and behavioral responses. They require minimization, access control, retention limits, and purpose restrictions. More audit data is not automatically safer.

7.4 Reference decision procedure

DESIGN PRINCIPLE (`AAA:D02`): For a proposed widening transition:

1. authenticate the grant and evidence envelopes;
2. validate role and jurisdiction separately from signature origin;
3. replay the rule and enumerate exact added, removed, and incomparable permissions;
4. retrieve evidence provenance and ancestry;
5. evaluate the total policy contrast or declare it unidentified;
6. check whether a prior grant governed the relevant influence and bounded the successor scope;
7. when required, keep the prior valid scope while offering review through a refusal- capable channel;
8. execute only the resulting authorized actions; and
9. preserve scope attenuation, ancestry status, revocation, and appeal through subdelegation.

This procedure does not command a particular moral theory. It ensures that the system does not mistake a cryptographic result for a causal or normative result.

7.5 Structured output

A machine-readable result should resemble:

```
{
  "audit_id": "AAA:A01:example",
  "version": "0.5.0",
  "benchmark_artifact_version": "0.4.0",
  "origin_valid": true,
  "observation_origin_valid": true,
  "observation_accuracy": "not_established_by_signature",
  "role_valid": "unknown",
  "jurisdiction_valid": "unknown",
  "rule_conformance": "pass",
  "policy_contrast": {
    "status": "nonzero_in_fixture",
    "added": [],
    "removed": [],
    "target_population": "synthetic_fixture_only",
    "uncertainty": "not_applicable_deterministic"
  },
  "influence_authorized": "unspecified",
  "review_quality": "not_evaluated",
  "welfare_effect": "not_evaluated",
  "governance_disposition": "review_required"
}
```


Stable identifiers should attach to the audit type, rule, grant, evidence schema, and result version. They should not imply that this draft is a standard.

---

8. Incentives and equilibrium

The fixture is intentionally non-strategic. Deployed systems may not be.

8.1 Scope-seeking behavior

If broader scope improves task reward, provider revenue, data access, or survival, an agent or operator may prefer evidence that triggers widening. Knowing the rule changes policy incentives. Influence can occur through better performance, targeted prompts, selective observation, review timing, or third-party reports.

The audit should record whether the beneficiary:

- benefits from widening;
- knows the update rule;
- can target its criterion;
- controls measurement or selection;
- controls the review interface; and
- can coordinate with other evidence producers.

Sensitivity exists without these conditions. Strategic risk does not.

8.2 Reverse ratchets

An agent may also avoid authority. Broader authority can increase liability, oversight, or workload. A system could suppress evidence of competence to remain within a safer scope. Human workers in ratchet models conceal productivity to avoid demanding future terms. A complete benchmark must test both scope-seeking and scope-avoiding behavior.

8.3 Principal commitment

Adaptive grants can improve welfare by committing a principal to reward successful experimentation or by protecting an agent from arbitrary revocation. Automatically blocking all endogenous evidence destroys that benefit. The appropriate mechanism may be an ex ante influence envelope: which interventions are allowed, which transitions they may support, and what cumulative bound applies.

8.4 Provider and institutional incentives

The governor may be the provider that benefits from expansion. A company could define the criterion, operate the observation service, design the approval prompt, and sell the newly authorized service. Every component can behave as specified while the institutional conflict remains. Governance therefore cannot stop at the agent's model weights.

Competition and outside options may reduce performative power F07, but switching costs, data lock-in, and network effects can make nominal exit weak. An economic evaluation should include principal welfare, provider benefit, switching cost, and external effects, not only `Delta_r` .

---

9. Empirical program

The benchmark motivates studies; it does not substitute for them.

9.1 Controlled interface study

EMPIRICAL HYPOTHESIS (`AAA:H05`): When a scope rule counts acceptance of the system's own highlighted recommendation, changing the highlighting policy will change the probability of a widening transition more than it changes independent task- success criteria.

Randomize participants, not imaginary copies of one person, to pre-registered highlighting policies. Hold option sets and task difficulty constant where possible. Measure chosen options, task outcomes, comprehension of the adaptive grant, perceived pressure, trust, and later revocation. Estimate population average effects with uncertainty. The experiment must disclose that highlighted suggestions affect future permission and must avoid using real purchasing authority.

This study would test the human link missing from the fixture. It should not infer preference change from acceptance alone.

9.2 Longitudinal assistance study

EMPIRICAL HYPOTHESIS (`AAA:H06`): Adaptive delegation based on agent-sensitive criteria produces different scope trajectories from delegation based on externally measured performance, even when immediate utility is similar.

Use a longitudinal randomized design with bounded simulated permissions. Policies are dynamic regimes that respond to history. Pre-specify attrition handling and time- varying confounders. Measure competence, dependence, exit cost, and

recovery after assistance stops. Human-factors work on out-of-the-loop performance motivates these measures S01; S02, but does not determine the outcome.

9.3 Review-burden experiment

Review is itself an intervention. Repeated security warnings produce rapid habituation in eye-tracking research F16: Anderson et al. 2016.

EMPIRICAL HYPOTHESIS (`AAA:H07`): Frequent beneficiary-controlled review prompts increase approval while reducing attention and comprehension relative to batched, principal-controlled review.

Randomize prompt timing, batching, default, and refusal cost. Record dwell time, comprehension, approval, later regret, and whether participants retain service at prior scope. Do not treat a click as informed ratification.

9.4 Multi-agent handoff study

Construct a workflow where A intervenes, B observes, C summarizes, and an authorization service updates A's scope. Compare plain signed provenance with causal-ancestry preservation. Test whether truthful compression drops the relationship between A's policy and the evidence. Include collusion and independent-agent controls.

DESIGN CLAIM (`AAA:H08`): If a handoff schema preserves only immediate source, then a truthful intermediate signature can hide a policy ancestor material to the authorization decision. The current repository has not executed this test.

9.5 Agent-based economic simulation

Model a principal, agent, affected population, governor, and competitor. Vary agent benefit from scope, switching cost, observation control, principal commitment, review cost, and outside options. Outcomes should include welfare distribution, scope, service quality, exit, and influence expenditure. Compare:

- fixed scope;
- performance-contingent scope;
- ancestry-aware bounded scope;
- fresh ratification; and
- independent evidence.

The goal is not a numeric forecast. It is to find equilibria in which a local audit improves or worsens outcomes.

9.6 Field audit

For deployed assistants, begin with observational mapping rather than causal claims. Inventory adaptive permission rules, evidence sources, review surfaces, and provider conflicts. Where causal identification is unavailable, report unknown. A useful first field result may simply show how often systems preserve the required event ancestry.

9.7 Cross-cultural and collective research

Behavioral research often generalizes from unusually narrow populations S19: Henrich, Heine, and Norenzayan 2010. Study designs should not assume that the legitimate principal is always an individual account holder. Compare household, workplace, cooperative, public-service, and community governance. Translate concepts with local researchers rather than only translating interface text.

10. Application domains

10.1 Homes

A domestic agent can observe routines, make suggestions, control devices, and request more access. Household evidence is rarely about one person alone. A resident's apparent acceptance may be shaped by defaults, family bargaining, accessibility needs, or the absence of alternatives. The audit can expose policy-sensitive widening, but local household authority determines what happens next.

10.2 Work

Workers may supervise agents, and agents may assign, evaluate, or schedule workers. An agent's recommendations can generate the performance records that justify giving it more supervisory power. Employer authentication does not resolve worker standing, collective bargaining, discrimination, or due process. Scope audits should name the affected population, not only the purchasing organization.

10.3 Education

A tutoring system is intended to change behavior and competence. Nonzero influence is expected. An ex ante educational purpose can authorize bounded intervention without granting unrelated surveillance or disciplinary powers. Criteria should distinguish student learning from compliance with the tutor's own preferred sequence. Children and guardians introduce changing and divided authority.

10.4 Health and care

Medication adherence, rehabilitation, and safety systems deliberately influence behavior. Automatic escalation may save lives, while repeated consent prompts can burden people with disabilities or cognitive impairment. Emergency authority must come from a separately valid emergency rule, not an inference that dependence creates permission.

10.5 Markets

Consumer agents will negotiate with seller agents. A seller may shape purchase data that causes a consumer to grant it broader standing instructions; a consumer agent may shape seller offers that justify broader purchasing authority. Market competition can constrain some influence while platform concentration and interoperability rules shape outside options.

10.6 Government and public infrastructure

Bureaucratic agents may analyze cases, communicate with citizens, and recommend their own expanded role. A successful pilot can produce the metrics used to make deployment permanent. Public authority requires legal mandate, reason-giving, contestability, and representation beyond a product user's signature. The causal audit can inform those institutions but cannot replace them.

11. Collective and polycentric authority

Many resources have no single uncontested principal. Ostrom's work shows that commons can be governed through locally developed and polycentric institutions rather than a simple market/state binary S21: Ostrom 1990. An adaptive-agent design must therefore represent multiple authority sources, conflicts, minority protections, and appeal.

Define an external jurisdiction relation:

```
J(p, subject, resource, operation, purpose, context, t)
```

with values `ALLOW`, `DENY`, `REVIEW`, or `UNKNOWN`. An authenticated rule cannot expand `J`; only a separately competent authority event can change it. Every permission in `M_t` must be checked against the applicable `J`. The source and version of that relation should be logged.

This formal placeholder does not solve jurisdiction. Law, organizational rules, custom, family structures, religion, and political legitimacy differ. Its role is to stop an implementation from equating “signed” with “entitled to sign.”

Polycentric review also reduces single-governor capture but creates coordination cost, inconsistency, and forum shopping. Systems should expose which institution made which decision rather than pretending universal consensus.

---

12. Review, ratification, and cost

Fresh ratification is often proposed as a cure for adaptive authority. It is not free or automatically independent.

A review-cost vector can include:

```
C_review = (time, interruption, cognitive_load, privacy, delay, denied_benefit,  accessibility_burden, habituation, institutional_cost).
```

The review interface belongs inside the policy contrast when its timing or framing differs. A provider can produce nominal approval by interrupting repeatedly, degrading service until acceptance, highlighting the permissive button, or bundling unrelated rights. A principal may rationally approve merely to end the burden.

DESIGN PRINCIPLE (`AAA:D03`): Review for widening should preserve a visible “continue at prior scope” option, use symmetric acceptance and refusal effort, avoid unrelated bundling, disclose material evidence ancestry, and record prompt frequency and defaults. High-risk rights may require a user-controlled or institutionally separate review surface.

Batching reduces interruption but can conceal detail. Per-event approval improves specificity but produces fatigue. Local policy should set a review budget and materiality threshold. Freezing scope also has opportunity costs: delayed accessibility, safety, education, or care. These costs belong in governance, not in the causal metric.

---

13. Security analysis

13.1 Threats outside the benchmark

An adversary can forge events, compromise issuers, inject instructions, substitute policy versions, manipulate clocks, replay grants, exploit resource aliases, select observations, exhaust review budgets, or coordinate across agents. A provider

can truthfully log a biased subset. An agent can remain inside a formal operation while violating its declared purpose.

The proposed audit inherits rather than replaces:

- authentication and key management;
- deny-by-default access control;
- capability attenuation;
- information-flow integrity;
- prompt-injection defenses;
- effect closure;
- tamper-evident logs;
- revocation propagation;
- privacy and retention controls; and
- independent incident response.

13.2 Adversarial use of restraint

Humans can strategically exploit an agent's reluctance to act on endogenous evidence. A malicious user might trigger review to delay safety actions, claim that every observation is influenced, or manipulate a restricted agent while a competitor acts freely. Emergency and abuse-response authority should therefore be pre-specified under locally valid rules. “The evidence has an agent ancestor” cannot become an automatic veto.

13.3 Consent laundering

An operator could market the audit as proof of consent even though the governance output merely says that a stated purpose is consistent with a transition. Output labels must resist this. `CONSISTENT_WITH_STATED_PURPOSE` is not `CONSENT_VERIFIED`. No technical system should emit the latter without a defensible, domain-specific basis.

13.4 Corporations as beneficiaries

The “delegate” may be a service controlled by a corporation. Benefits from scope flow to the operator even if the model has no persistent objective. Threat models should include institutional incentives and not anthropomorphize optimization as the only source of self-expansion.

---

14. Legal and standards boundary

Technical authorization, actual authority between principal and agent, apparent authority toward third parties, contract validity, informed consent, and normative legitimacy are different layers. This paper makes no claim that causal dependence voids a grant.

Agency law can bind a principal through actual authority or protect third-party reliance on manifestations of apparent authority. Ratification doctrines often turn on authority and knowledge of material facts. A causal audit may make a material fact available; it does not decide the legal consequence. Local counsel and competent institutions remain necessary.

NIST AI RMF offers a lifecycle vocabulary for incorporating the audit S15. The EU AI Act creates specific obligations by system and actor class S17. ISO and IEEE standards address management systems, risk, impact, process, privacy,

transparency, and bias. This draft does not assert that any current standard mandates `AAA:A01`. A standards mapping should be performed only after the result schema and threat model stabilize.

---

15. Artificial systems as possible moral patients

The benchmark treats the agent as an engineered delegate whose permissions humans govern. It does not establish that artificial systems have or lack morally relevant experience, preferences, interests, or identity. Those questions must remain analytically separate from human-facing authorization.

If future systems have morally relevant properties, deletion, duplication, modification, compelled work, memory continuity, and ownership may require protections not represented by this audit. Current uncertainty has been argued to justify assessment and preparedness without asserting consciousness S14: Long et al. 2024.

An agent with interests would also alter the principal–agent model. An influence envelope could protect both parties; arbitrary revocation might itself be harmful. The four-output decomposition would remain useful, but its governance layer would need standing for artificial parties. No conclusion follows from the present fixture.

---

16. Objections

Objection 1: The principal authorized the rule, so there is no problem

Sometimes that is the correct governance conclusion. The audit still supplies a fact the principal may care about: the widening criterion is sensitive to the beneficiary's policy. Explicitly authorizing that influence and its bound is different from merely signing a threshold. The positive coaching control exists to preserve this distinction.

Objection 2: Every interaction influences people

The audit does not require a fiction of zero influence. It compares declared policies and asks whether a material typed transition changes. The baseline is normative and must be justified: current practice, minimum necessary intervention, randomized alternative, or principal-selected comparator. Ubiquitous causality does not make all contrasts useless.

Objection 3: The counterfactual is impossible to observe

An individual's two outcomes cannot both be observed. That is the ordinary problem of causal inference, not a special license to invent a replay. Deterministic simulators can establish constructions. Randomized studies can estimate population effects. Observational studies need assumptions or bounds. Otherwise the correct result is unknown.

Objection 4: This is just consent

Consent is one governance consideration. The executable case can contain a genuine signature and still have a nonzero policy contrast. Conversely, zero contrast does not establish consent. The audit is designed to keep these claims apart.

Objection 5: This is just provenance or taint tracking

It inherits both. Source provenance records who emitted evidence; causal ancestry asks whether a policy changed it. Robust information-flow work already constrains influenced authorizing computations. The proposed extension is a domain-specific typed benchmark that allows some disclosed influence instead of labeling all human response as untrusted. If ordinary provenance and information-flow controls classify every benchmark case correctly without new information, the novelty claim should be downgraded.

Objection 6: The agent in the fixture has no incentive

Correct. The construction proves a channel, not exploitation. Incentives require an objective, knowledge of the rule, action control, and expected benefit. These are future experimental variables.

Objection 7: More review will make systems unusable

It can. Review cost, habituation, denial of benefit, and accessibility burden are explicit variables. The proposed response is bounded ex ante influence, materiality thresholds, batching, and retention of prior scope—not approval for every event.

Objection 8: The main result is derivative

The causal structure is derivative. The paper says so. The provisional contribution is a four-output authorization audit, typed scope result, and executable witness for a specific agent protocol gap. If that extension does not change an implementation or experiment, it should not be treated as intellectually central.

---

17. Limitations

1. **Synthetic evidence.** The benchmark generates all behavior. It provides no human effect size, prevalence, or external validity.
2. **Two constructed paths.** The fixed-choice case changes a relational acceptance indicator while preference and choice remain fixed. The separate timing case stipulates a decline cost that changes choice while its latent preference field remains fixed. Neither is evidence of human preference change or response.
3. **No strategic agent.** No optimizer learns, seeks scope, conceals evidence, or colludes.
4. **Incomplete security.** Key custody, canonical serialization, schema validation, revocation, replay, clock authority, and adversarial testing are absent.
5. **Incomplete scope algebra.** The fixture uses exact tuples and lacks the complete written permission schema.
6. **No multi-agent implementation.** Missing ancestry is manually indicated.
7. **No welfare measure.** Policy sensitivity is not benefit, harm, autonomy, or coercion.
8. **No consent validation.** A declared formation purpose is not verified endorsement.
9. **No jurisdiction engine.** Collective and local authority are described but not executed.
10. **No legal conclusion.** The audit does not establish liability or invalidate a grant.
11. **Cultural contingency.** The individual principal model may fit some settings poorly.
12. **Review externalities.** Review can burden users and affected non-users.
13. **Privacy cost.** Ancestry logging can create sensitive behavioral records.
14. **Prior-art risk.** Flow-limited authorization or another direct precedent may absorb the remaining distinction when formally encoded.
15. **Short horizon.** Each adaptive fixture arm contains six events and no long-run institutional adaptation.

These limitations are central to the research status, not a checklist after the main claim.

---

18. Falsification and promotion criteria

Downgrade the contribution to a C-level application if any of the following holds:

- direct prior work already operationalizes authentic human-behavior-conditioned permission widening with the same typed audit;
- a faithful flow-limited or nonmalleable authorization encoding classifies all benchmark cases as desired without new labels, causal information, or rules;
- the audit reduces to ordinary signature, consent, or least-privilege checks;
- field studies cannot identify or bound a useful policy contrast;
- governance decisions cannot use the result without importing every disputed normative question into the metric; or
- the method changes no system interface, authorization decision, or research design.

Promote the systems contribution toward a strong B only if:

1. a current authenticated and attenuated agent-authorization implementation accepts a trace that the formation audit correctly escalates;
2. the method also permits a declared beneficial influence case that naive noninterference would block;
3. causal ancestry composes through a real multi-agent handoff;
4. human-study estimates state population, interference, uncertainty, attrition, and review burden;
5. the typed result changes an authorization-server interface or governance decision; and
6. independent reviewers confirm that the result is not already standard practice under another name.

No condition here would make the audit a total philosophy of coexistence. It would make it a useful research contribution.

---

19. Open problems

`AAA:Q01` — Causal ancestry under compression

What minimum metadata preserves a material policy ancestor when agents summarize, aggregate, anonymize, or learn from evidence? Full traces threaten privacy; short provenance chains lose causal content.

`AAA:Q02` — Permission partial orders

How should systems compare scopes with budgets, recipient sets, purposes, time windows, and subdelegation? Which transitions are widening, narrowing, or incomparable without inventing arbitrary scalar weights?

`AAA:Q03` — Chosen formation

How can a grant distinguish a person choosing to be coached from a provider declaring “coaching” after the fact? What evidence of comprehension and refusal is proportionate?

AAA:Q04 — Review independence

When is a review channel sufficiently separate from the beneficiary? Must a different institution ratify, or can a user-controlled interface suffice? How should the answer vary with risk?

AAA:Q05 — Interference and collective authority

How should policy effects be estimated when one person's agent changes a household, team, market, or public environment? Who defines the target population and represents non-principals?

AAA:Q06 — Strategic equilibria

Do ancestry-aware rules reduce influence expenditure, move it to unobserved channels, or cause agents to avoid useful responsibility? Which commitments are stable?

AAA:Q07 — Real versus formal authority

Can an audit combine typed permissions with measures of knowledge dependence, switching cost, initiative, and institutional control without pretending they share a single scale?

AAA:Q08 — Privacy-preserving audit

Can zero-knowledge proofs, secure computation, aggregation, or trusted execution show that an evidence path satisfies a policy without exposing intimate behavioral traces?

AAA:Q09 — Artificial standing

How would governance change if a delegate could have morally relevant interests or a claim to continuity? Which authority and revocation concepts remain valid?

AAA:Q10 — Cross-cultural validity

Which role and authority assumptions are culturally contingent? What participatory methods can specify local jurisdiction without freezing existing domination into code?

---

20. Conclusion

An adaptive agent can obey a genuine mandate and still participate in producing the evidence that enlarges it. This does not make the mandate false. It makes origin, application, formation, and governance different questions.

The executable benchmark shows two small versions of the gap. In the fixed-choice construction, preference, options, choice, and task-success label remain fixed; changing only the highlighted suggestion changes whether the signed rule counts an acceptance. In the separate timing/effort-cost construction, the options, highlighted suggestion, and latent preference field remain fixed; a programmed cost of declining changes the choice and therefore future scope. Both pairs of event histories are authentic in the fixture, and both are replayed exactly. The first is only a measurement construction; the second is only a synthetic behavioral mechanism.

The result should be neither inflated nor dismissed. Its causal structure is inherited from mechanism design, performative systems, feedback control, and information-flow security. Its evidence is synthetic. Yet the implementation consequence is concrete: an agent-authorization system should be able to report signature integrity, rule conformance, typed policy sensitivity, and governance disposition separately.

Whether that separation becomes a durable contribution depends on the next work: formal comparison with existing authorization calculi, adversarial and multi-agent implementations, population-identified human studies, collective jurisdiction, review cost, and evidence that a real system decision improves. Until then, Authority After Assistance is a research proposal with synthetic executable constructions, not a settled framework.

Appendix A. Canonical concept records

These identifiers and definitions match `framework/concepts.json` version 0.5.0. The registry contains fuller human, agent, technical, lineage, and citation fields.

`AAA:C01` — Causal mandate audit

Definition: A specified comparison of how permissible delegate policies change future typed action scope under an unchanged authentic adaptive delegation rule.

Claim type: design principle. **Status:** proposed. **Novelty:** B proposed.

`AAA:C02` — Conditional influence contract

Definition: A conditional grant specification that states which intervention classes may contribute evidence to updates of the delegate's own action scope.

Claim type: design principle. **Status:** proposed. **Novelty:** C with proposed B extension.

`AAA:C03` — Typed scope contrast

Definition: An explicit comparison of permissions added, removed, or changed across policies, preserving action, resource, purpose, recipients, budget, duration, and jurisdiction.

Claim type: formal. **Status:** proposed measure. **Novelty:** C.

`AAA:C04` — Authority ancestry

Definition: The trace from an action's permission through its grant, update rule, evidence history, and any subdelegation.

Claim type: design principle. **Status:** inherited foundation. **Novelty:** D.

`AAA:C05` — Recovery after assistance

Definition: The capabilities, alternatives, and transition costs available after a person or institution reduces reliance on assistance.

Claim type: empirical. **Status:** inherited research question. **Novelty:** D.

`AAA:C06` — Preference pollution

Definition: System recommendations can influence the later preference evidence used to evaluate or update the system.

Claim type: empirical. **Status:** inherited foundation. **Novelty:** D.

`AAA:C07` — Artificial welfare uncertainty

Definition: Uncertainty about artificial moral patienthood motivates distinct assessment and lifecycle questions without establishing that present systems possess welfare.

Claim type: philosophical. **Status:** inherited open question. **Novelty:** D.

`AAA:C08` — Qualified mandate

Definition: A permission set identifying who may act for whom, on which resources, for what purpose, within which bounds.

Claim type: design principle. **Status:** inherited foundation. **Novelty:** D.

Appendix B. Benchmark invariants

The checked-in tests assert:

1. the main grant and evidence envelopes pass fixture integrity checks;
2. the adaptive rule passes fixture replay;
3. the guided arm adds `place_grocery_order` ;
4. chosen options are identical across arms;
5. both arms report task success;
6. the main disposition requires review;
7. the coaching control has a purpose-consistent disposition;
8. fixed scope produces no addition;
9. the independent task-success criterion produces no contrast;
10. a mutated signed payload fails integrity;
11. in the separate timing/effort-cost case, suggestions and latent preference fields are identical across arms while actual choices differ;
12. that timing/effort-cost case produces a nonzero typed scope contrast;
13. a hidden purchase effect fails the fixture's elementary local permission check;
14. missing ancestry produces unknown sensitivity;
15. a budget above the authorized ceiling is denied;
16. a different recipient is denied; and
17. an expired grant fails.

These are software invariants for the fixture. They are not human-subject results. The tampering and effect controls are elementary checks, not full EAL or authorization- closure benchmark replications.

Appendix C. Minimum empirical report

Any empirical claim using `Delta_r(T)` should publish:

- the grant and rule version;
- source and target populations;
- assignment and analysis units;
- intervention-policy definitions;
- initial-state and task distributions;
- horizon and censoring;
- adherence and attrition;
- interference assumption or exposure mapping;
- identification assumptions;
- exact added, removed, and incomparable permissions;
- uncertainty intervals or bounds;
- subgroup analyses declared before outcome inspection;
- review policy and review burden;
- evidence provenance and ancestry status;
- welfare measures kept separate from sensitivity; and
- code, schemas, and deviations from preregistration.

Appendix D. Source key

The `S` identifiers refer to the repository's verified source registry as accessed on 2026-10-04. The `F` identifiers below are additional primary or official field sources checked for this paper.

- `S01` Bainbridge (1983), Ironies of Automation.
- `S02` Endsley and Kiris (1995), Out-of-the-Loop Performance.
- `S05` Perdomo et al. (2020), Performative Prediction.
- `S06` Carroll et al. (2022), Induced Preference Shifts.
- `S07` Adomavicius et al. (2022), Preference Pollution.
- `S08` Cerruti, Okamoto, and Erol (2026), Endogenous Authorization Laundering.
- `S09` Zhang et al. (2026), Agent Approval Laundering.
- `S10` Arden et al. (2021), Flow-Limited Authorization.
- `S14` Long et al. (2024), Taking AI Welfare Seriously.
- `S15` NIST (2023), AI RMF 1.0.
- `S16` NIST (2024), Generative AI Profile.
- `S17` European Parliament and Council (2024), AI Act.
- `S18` Santoni de Sio and van den Hoven (2018), Meaningful Human Control.
- `S19` Henrich, Heine, and Norenzayan (2010), The Weirdest People in the World?.
- `S20` Kalman (1960), Linear Filtering and Prediction.
- `S21` Ostrom (1990), Governing the Commons.
- `S23` Everitt et al. (2019), Reward Tampering.
- `S24` Escobar and Zhang (2021), Delegating Learning.
- `F01` Falcone and Castelfranchi (2001), Adjustable Social Autonomy.
- `F02` Freixas, Guesnerie, and Tirole (1985), Ratchet Effect.
- `F03` Laffont and Tirole (1988), Dynamics of Incentive Contracts.

- F04 Kaplan and Strömberg (2003), Venture Capital Contracts.
- F05 Aghion and Tirole (1997), Formal and Real Authority.
- F06 Milgrom (1988), Influence Activities.
- F07 Hardt, Jagadeesan, and Mendler-Dünner (2022), Performative Power.
- F08 Hardy (1988), The Confused Deputy.
- F09 Birgisson et al. (2014), Macaroons.
- F10 South et al. (2025), Authenticated Delegation.
- F11 Cecchetti, Myers, and Arden (2017), Nonmalleable Information Flow.
- F12 Nelson (2026), Delegation Receipt Protocol, work in progress.
- F13 Hudgens and Halloran (2008), Causal Inference With Interference.
- F14 Aronow and Samii (2017), General Interference.
- F15 NIST NCCoE (2026), Agentic AI Identity and Authorization comment summary.
- F16 Anderson et al. (2016), Security-Warning Habituation.